Analysis choices and orbital sampling shape a particle-defined South Atlantic Anomaly footprint

A January 2024 NOAA POES / EUMETSAT MetOp sensitivity case study

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Abstract. Locations and extents of the South Atlantic Anomaly (SAA) derived from low-Earth-orbit particle data are outputs of a mapping pipeline, yet the pipeline's choices are rarely varied and reported together. This single-month methodological case study (January 2024) maps NOAA/NCEI SEM-2 MEPED omnidirectional differential proton fluxes (nominal 25, 50 and 100 MeV) from five NOAA and MetOp spacecraft, defines a candidate high-flux footprint as the cells at or above a percentile of a coverage-gated latitude–longitude grid, and summarizes it by spherical area and a flux-and-area-weighted centroid. The threshold redefines the object: tightening it from the top 20% to the top 1% moves the centroid by 351–433 km, while the area contraction is almost entirely prescribed by the percentile rule. Sampling support must be separated from field differences: a three-day map agrees with the monthly map at an area-weighted Jaccard index of 0.540 (138 km) end to end, but at 0.829 (14 km) when both footprints are recomputed on the cells eligible in both. Energy channel matters: the 100 MeV (p3) top-10% centroid lies 210.4 km south-east of the 25 MeV (p1) centroid, a direction that persisted under paired calendar-block resampling. A single record can dominate an inter-satellite comparison: after a stated post-hoc quality rule excludes one NOAA-15 record whose >35 MeV count rate exceeds its own >16 MeV rate, the five spacecraft centroids lie within 113.4 km of one another. These are sensitivities of one month's data, not a validated physical model of the anomaly.

1 Introduction

The South Atlantic Anomaly is the region where the inner proton belt reaches its lowest altitudes over South America and the South Atlantic, so that satellites in low Earth orbit encounter elevated energetic-particle fluxes there. Its location and extent are commonly described from particle measurements rather than from the magnetic field alone. Long particle records have been used to follow the SAA's slow westward drift and its solar-cycle modulation in location, area and peak intensity: from the particle monitor on the Rossi X-ray Timing Explorer (Fürst et al., 2009), from particle-noise pulses in an ultraviolet photometer (Schaefer et al., 2016), from NOAA-15 proton data fitted with Gaussian forms (Qin et al., 2014), and from NOAA/MetOp SEM-2 proton fluxes over 1998–2022 (Ginisty et al., 2024). Shorter-term changes have been mapped by pooling several NOAA and MetOp spacecraft; Xu et al. (2025), for example, used a three-day running average of integral proton channels to resolve a storm-time decrease that month-long averages would smooth.

Each of these descriptions is an estimate produced by a pipeline. A particle-defined "SAA region" depends on how flux is gridded, what counts as high flux, which energy channel and cell statistic are used, how long data are accumulated, and which spacecraft sampled the region. These choices are usually fixed once and reported in passing. When they are varied at all, the variation is typically one factor at a time. It is therefore hard to know how much of a reported location or area belongs to the particle environment and how much to the analysis.

This paper asks a narrower question than SAA physics. **For one month of public NOAA/NCEI SEM-2 data, how much do the location and extent of a percentile-defined candidate high-flux footprint change under controlled changes of threshold, cell statistic and grid, proton channel, accumulation window and spacecraft, and which of those changes are explained by sampling support or by individual observations rather than by the flux field?** January 2024 is used as a single, documented case.

The contribution is methodological rather than physical: a reproducible way of measuring and reporting how much of a particle-defined SAA location and extent belongs to the analysis. It has three parts. First, we define the footprint estimands (selected set, spherical area and coordinate-moment centroid) and their denominators explicitly, and we separate the parts of each comparison that the definition itself prescribes. Second, we compare short-window maps with the monthly map in two ways: end to end, using the original footprints, and on shared eligible support, after recomputing each map's percentile on the cells both maps can support. Third, we trace a large inter-satellite difference to one source record, check it against the provider's count-rate file, and exclude it under a stated post-hoc quality rule while reporting the unfiltered result alongside. Channel differences are

examined with a paired calendar-block resampling protocol fixed before production resampling. All values in this paper are generated from a machine-readable claims register tied to hashed outputs, and the complete computation is reproducible from the raw daily files.

The results are case-study sensitivities, not a validated physical model of the SAA's position or extent. They do not define a true SAA boundary or centre, do not measure dose or hazard, and none of the displacements reported is a measured motion of the anomaly. Because they come from one month, they show which choices can matter and by how much in that month; what is meant to carry over to other periods is the procedure, not the magnitudes.

2 Data and methods

2.1 Product and provenance

We use the processed SEM-2 daily NetCDF files distributed by NOAA's National Centers for Environmental Information (NOAA National Centers for Environmental Information, 2024) for all 31 days of January 2024 from NOAA-15, NOAA-18, NOAA-19, MetOp-01 and MetOp-03 (155 files; SHA-256 manifest in Supplement S1). The archive directory is named `l1b/v01r00`. The files themselves declare `processing_level = "Level 2, calibrated fluxes"`, consistent with the NOAA processing description, in which level-1B sensor data are converted to physical units and supplemented with field-model parameters (Green, 2013). We therefore describe the product by its file metadata, not by the directory name. The file title also states that the data have known contamination problems.

The analysed variables are `mep_omni_flux_p1`, `p2` and `p3`. According to the OMNI flux algorithm description (Machol, 2012), count rates from four overlapping omnidirectional channels (16–35, 35–250, 70–250 and 140–250 MeV) are redistributed into non-overlapping bands. A piecewise power law in energy is then fitted, and the fitted differential flux is evaluated at 25, 50 and 100 MeV (p1, p2, p3). The file metadata give these variables as differential proton flux at 25, 50 and 100 MeV in `#/cm2-s-str-MeV`. The user manual's variable table prints the unit without the per-energy factor (Green, 2013). We follow the algorithm definition and the file metadata and report flux in $cm^{-2}\ s^{-1}\ sr^{-1}\ MeV^{-1}$. The three channels are therefore **fitted outputs from shared counts and a shared spectral model**, not independent instruments. NOAA's calibration and processing are inputs to this study; we perform no cross-satellite or cross-channel intensity calibration of our own.

2.2 Quality handling and record accounting

Records are restricted to latitude −70° to 20° and longitude −100° to 20° (longitudes converted from [0°, 360°) to [−180°, 180°)). Rows with the in-flight-calibration flag `mep_IFC_on = 1` are removed (1 NOAA-15 regional record; none on the other spacecraft). The flag's documented values are 0 and 1, yet most regional records carry −1 (e.g. 192,334 of 205,153 NOAA-19 records). Its meaning is not documented, so these records are retained and left uninterpreted. Following the algorithm's fit-flag definition (−1 = no fit performed) and the documented minimum of zero for the OMNI fluxes (Green, 2013; Machol, 2012), a channel value is set missing when it is negative or its fit flag is −1. Valid zeros and fit types 0–2 are retained; no upper cutoff or automatic despiking is applied. In the study region this affects 29 p1 values (25 NOAA-15, 4 NOAA-18, none on the other spacecraft). Missing values are never interpolated or zero-filled. Supplement S2 gives the per-satellite accounting.

Only the 31 NOAA-19 files were cached with recorded hashes before this analysis was finalized. The other 124 January files were retrieved again from the same archive, so byte hashes of the files originally used for some earlier project checkpoints are not available. Regional tables rebuilt from the retrieved files reproduce the saved values exactly, which supports value lineage but does not recover the unrecorded historical bytes.

Post-hoc quality rule QC-1. The OMNI algorithm fits separate power laws over four energy ranges (fit flag 0) when the count rates allow it. Otherwise it falls back to a single power law built from the two lowest channels (flag 2) or from one anchor with the default slope $\gamma = -2.9$ (flag 1) (Machol, 2012). This fallback is a documented fit mode and is common at low count rates, so we do not treat it as invalid. After the satellite comparison had drawn attention to one NOAA-15 record (Sect. 2.8), we adopted the following rule and applied it to every native record of the 155 files. A record's OMNI values are set missing when (i) its fit is a fallback (flag 1 or 2); (ii) its 10 preceding and 10 following native records are contiguous in time (every step positive and at most 6 s, three nominal cadences) and all received the regular fit; and (iii) for at least one channel, all neighbour values

are finite and positive and the record's value exceeds 100 times the median of the preceding records and 100 times the median of the following records. The three channel values of a flagged record come from one fit and are excluded together. Values are never clipped or replaced, records that cannot be evaluated are kept, and the source files are not altered.

Condition (ii) restricts the rule to stretches where the counts supported the regular fit, which the algorithm attempts only when the summed raw count rate is at least 25 s^{-1} (Machol, 2012). There, record-to-record variation is small: across 349,822 evaluable regular-fit records, no p1 value departs from both neighbourhoods by more than a factor of 3.3. The documented fallback itself departs further, because one power law with a fixed slope does not follow the local spectrum: in p1, the 1,242 isolated fallback records sit a median factor of 3.1–5.2 above their regular-fit neighbours (by satellite), and at most 19.8. The ratio must exceed that range, or the rule would reject a documented fit mode for its known bias; 100 is 5.1 times its upper end. Neither the NOAA documentation nor the literature we found gives an automatic criterion for single-record departures: the provider's quality assessment is flag-based (Machol, 2012), and a cross-calibration of the same >16 MeV omnidirectional data removed scattered erroneous points by manual inspection (Lin et al., 2019). QC-1 is therefore a post-hoc decision rule, stated in full so that it can be replayed. In a perturbation study (Supplement S7), 26 of 35 parameter variants give exactly the same excluded set, including every ratio from 20 to 2000. The set changes only when the ratio drops into the fallback's own departure range (4 records at 15×, 30 at 10×) or when condition (ii) is dropped (32 records, mostly low-flux fallback values beside near-zero neighbours). QC-1 excludes 1 record of 6,313,670 (Sect. 3.5); every other fallback value is retained.

2.3 Coverage-gated grids

Valid values are binned by sub-satellite latitude and longitude into 5° and 2° cells. The nominal grids have 432 and 2,700 cells. Each cell receives its sample count, mean and median. A cell is **eligible** for a monthly map when it has at least 30 valid samples. Shorter windows of d days use $\max(3, \mathrm{round}(30d/31))$ samples, i.e. 3, 3, 7, 14 and 30 for 1, 3, 7, 14 and 31 days. In January, 432 of 432 5° cells and 2,685 of 2,700 2° cells are eligible for NOAA-19 p1. Ineligible cells carry no statistic and never enter a footprint. We distinguish cells with **no valid observation**, cells **observed but below the sample minimum**, and cells **eligible but not selected**.

2.4 Footprint, area and centroid

Let E be the eligible cells of one map and F_i the gridded statistic (mean or median) of cell i. For a percentile $q \in \{80, 90, 95, 98, 99\}$ ("top 20%" … "top 1%"), the cutoff c_q is the linear-interpolation q-th percentile of $\{F_i : i \in E\}$, and the footprint is

$$S_q = \{i \in E : F_i \geq c_q\}. \tag{1}$$

The percentile is taken over eligible **cells**, not over equal areas or individual records. Cell i spanning latitudes φ_i^- to φ_i^+ and longitude width $\Delta\lambda$ (in radians) has spherical area $A_i = R^2 \Delta\lambda (\sin\varphi_i^+ - \sin\varphi_i^-)$ with $R = 6371$ km. The selected area is $A(S_q) = \sum_{i \in S_q} A_i$. The centroid is the coordinate moment

$$(\overline{\varphi}, \overline{\lambda}) = \frac{\sum_{i \in S_q} w_i (\varphi_i, \lambda_i)}{\sum_{i \in S_q} w_i}, \quad w_i = F_i A_i, \tag{2}$$

with cell-centre coordinates (φ_i, λ_i). It is a latitude–longitude moment, valid here because the region does not cross the antimeridian. It is not a three-dimensional barycentre, and its value depends on the chosen region. Distances between centroids are great-circle (haversine) distances. For two footprints X, Y on the **same aligned grid**, the area-weighted Jaccard index is

$$J(X, Y) = A(X \cap Y) / A(X \cup Y), \tag{3}$$

and it is undefined when both sets are empty. Cells of different resolutions are never compared as pixels.

2.5 Controlled comparisons

Table 1 summarizes the design. Each family changes one focal factor and crosses it with the same grid × statistic × threshold settings; factors are not combined into new joint experiments. The canonical configuration universe has 360 entries (20 threshold, 60 channel, 180 time-window and 100 satellite). They are not independent. The

NOAA-19 monthly p1 map appears in all four families, and the first calendar week is the same date range as the cumulative seven-day window. Only 280 of the 360 configurations are distinct (Supplement S3).

Element	Definition
Data	NOAA/NCEI processed SEM-2 daily files, January 2024; NOAA-15, −18, −19, MetOp-01, −03; `mep_omni_flux_p1..p3`
Region and quality	lat −70°…20°, lon −100°…20°; `mep_IFC_on = 1` removed; negative or fit = −1 values missing; post-hoc rule QC-1 removes fallback-fit values > 100× both regular-fit neighbourhoods (1 record)
Reference setting	NOAA-19, p1 (25 MeV), whole month, 5° cells, cell mean, top 10% ($q = 90$)
Eligible population	grid cells with ≥ 30 valid samples (month); $\max(3, \mathrm{round}(30d/31))$ for d-day windows
Footprint	eligible cells with statistic ≥ linear q-th percentile over eligible cells (Eq. 1)
Estimands	selected cells; spherical area (km²); flux × area weighted coordinate centroid (Eq. 2); great-circle distance (km); area-weighted Jaccard (Eq. 3); normalized mass overlap (Eq. 4)
Threshold family	top 20, 10, 5, 2, 1% × {5°, 2°} × {mean, median}; NOAA-19 p1, month
Channel family	p1, p2, p3 × the same 20 settings; paired 1/3/7-day calendar-block resampling, 1,000 replicates each, for the reference setting
Time family	1, 3, 7, 14 cumulative days from 1 January; four disjoint weeks; month reference; end-to-end and shared-support comparisons
Satellite family	five spacecraft × 20 settings; QC-1 applied (primary) and unfiltered (sensitivity)

Table 1: Design, reference setting, eligible population and estimands. All flux values are fitted differential fluxes in $\mathrm{cm^{-2}\ s^{-1}\ sr^{-1}\ MeV^{-1}}$; spacecraft and channels are compared by location and shape only.

2.6 Temporal coverage versus conditional morphology

A short-window footprint F_W can differ from the January footprint F_M because the field differs or because the window supports fewer cells. Let C_W and C_M be the eligible cells of window W and of January at the same grid, statistic and threshold. We report (i) the eligible area fraction of each window; (ii) the **end-to-end** comparison of the original footprints, $J(F_W, F_M)$, centroid distance and area ratio; and (iii) the **monthly-footprint observability** $O = A(F_M \cap C_W)/A(F_M)$, which measures how much of the monthly selected area meets the window's eligibility rule, not how well the footprints agree. (iv) On the pair's shared support $S = C_W \cap C_M$, we recompute **each** map's own percentile cutoff over S under the same convention and compare the resulting **conditional** footprints with J, centroid distance and area ratio. S is formed separately for each window–reference pair; no global intersection across windows is used, and masks are not simply intersected. Empty support, empty union or zero total weight return an explicit undefined value with a reason, never a zero or a perfect score. We also count distinct UTC days with valid data per cell as a description of sampling (not a filter; a record count is not a count of independent passes). The month contains every window, so all agreement with it is descriptive, not independent validation. These comparisons are descriptive; they do not attribute differences to sampling or to physical evolution.

2.7 Channel stability and continuous-field summaries

Channel footprints are compared by signed centroid differences (second channel minus first) and by distance. To describe how stable these differences are to day-to-day sampling, a protocol fixed before execution resamples calendar days in blocks of 1, 3 or 7 consecutive UTC days, with replacement and without wrap. Draws are concatenated and trimmed to 31 day-slots, 1,000 replicates per block length, seed 20260929 (PCG64), and the **same** day draws are used for all three channels. In every replicate, coverage, percentiles, footprints and centroids are recomputed. Moving-block resampling was developed for stationary dependent series (Künsch, 1989), and its consistency under some periodic non-stationarity has been studied for mean-type statistics (Synowiecki, 2007). Neither result covers a 31-day, coverage-gated, thresholded centroid. The quantiles are therefore reported as **conditional resampling stability**, not as calibrated confidence intervals for a physical position, and a positive distance quantile is not evidence of physical significance.

As a threshold-free complement, each channel's field on the common eligible support is normalized as

$$p_{c,i} = F_{c,i}A_i / \sum_j F_{c,j}A_j, \quad \mathrm{overlap}(c,d) = \sum_i \min\left(p_{c,i}, p_{d,i}\right), \tag{4}$$

which is invariant to any positive scale factor per channel. The overlap is a normalized mass overlap. It is not a footprint Jaccard index and not a "fraction of the SAA shared". The whole-region moment of p is a field

summary, not an SAA centre. It is a different statistic from the footprint centroid, and the two distances are not additive components of one displacement.

2.8 The influential NOAA-15 record

A review of the principal satellite comparison found that one NOAA-15 record (2024-01-19T09:14:30.051Z) strongly influenced its cell mean. Its OMNI values come from the one-point fallback fit with the default slope, while all its native neighbours received the regular fit, and its p1 value lies far above theirs. NOAA's level-1a companion file for that day, which holds the OMNI count rates from which the fluxes are fitted, shows why the value is an artefact rather than a flux enhancement: in this record the >35 MeV count rate is 539 times the simultaneous >16 MeV rate. Every proton above 35 MeV is also above 16 MeV, so the two rates cannot describe the same particle population, and the stored fluxes do not follow from the record's own >16 MeV measurement. The checks do not identify the instrumental or telemetry cause.

QC-1 (Sect. 2.2) was written after this record was found; the record satisfies it and is excluded from the primary analysis. Every setting is also recomputed without QC-1 and reported as the **unfiltered** sensitivity; the effect of the exclusion is described in Sect. 3.5. The record-level diagnostics (file identity and native index, neighbour statistics, the co-located directional channels, the spectral-ratio argument and the full count-rate evidence), the reproduction of an earlier exact-record omission and a historical triage run before QC-1 existed are given in Supplement S7.

2.9 Software

The pipeline uses Python with NumPy (Harris et al., 2020), pandas, xarray/netCDF4 and Matplotlib (Hunter, 2007); coastlines are from Natural Earth (Natural Earth, n.d.). AI coding assistants (OpenAI ChatGPT and Codex, Anthropic Claude) were used to write the analysis code and to carry out parts of the analysis under the author's direction. The author designed the study, chose the methods and quality rules, and reviewed the results; every reported number is regenerated by the repository code and read from the claims register. Code availability and reproduction are described in Section 6.

3 Results

3.1 Reference footprint

For the reference setting (Fig. 1a), the top-10% cutoff is 12.56 cm^{-2} s^{-1} sr^{-1} MeV^{-1}. The footprint contains 44 of 432 eligible 5° cells (12.61 × 10^6 km^2), and its centroid is at −20.89° latitude, −55.48° longitude. The highest cell mean (31.46) is in the adjacent cell centred at −22.5°, −52.5°. Every 5° cell meets the monthly sample minimum (Fig. 1b). At 2°, 15 observed cells fall below it and are left blank, and 0 cells lack observations entirely (Fig. 1c). The selected sector over South America and the South Atlantic persists in every setting examined below. What changes is its extent and the location of its summary centroid.

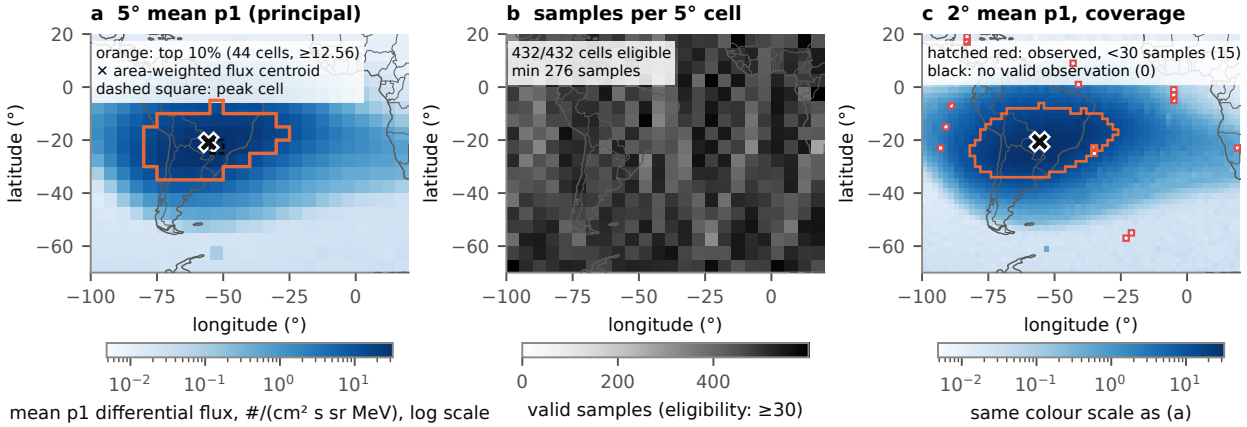


Figure 1: Reference map, January 2024, NOAA-19 p1. (a) Eligible 5° cell means (log colour scale). Orange: top-10% footprint (cells ≥ 90th percentile over eligible cells). Cross: area-weighted flux centroid (Eq. 2). Dashed square: cell with the highest mean. (b) Valid samples per 5° cell; all 432 meet the ≥ 30 rule. (c) The same field at 2° on the same colour scale. Hatched red cells are observed but have fewer than 30 samples (no statistic); 0 of the 2,700 2° cells lack observations entirely. The footprint is a percentile-defined analysis construct, not an SAA boundary.

3.2 Threshold and estimator

Tightening the percentile from top 20% to top 1% moves the reference centroid by 433 km, mostly southward (Fig. 2a). Across the four grid × statistic combinations the displacement is 351–433 km. The path is not monotone: the successive steps on the 5° mean grid are 134, 63, 69 and 205 km, and the last step selects only five cells. Mean and median grids follow similar, not identical, paths.

Area behaves differently. Selected area falls by a factor of 17.7 between top 20% and top 1% (20.1 at 2°), but the percentile rule already prescribes a nominal factor of 20 in selected **cells**. The realized cell-count ratio is 17.4, departing from 20 only through discreteness and ties. The area ratio departs from the cell ratio only because cell areas vary with latitude. Figure 2b shows the selected areas close to the line fixed by the nominal fraction. The area contraction is therefore a property of the definition, not an independent empirical finding. Nested-threshold overlap is likewise largely fixed by the nesting of percentile sets. The informative part of the threshold family is **where** the tighter sets fall, not how large they are.

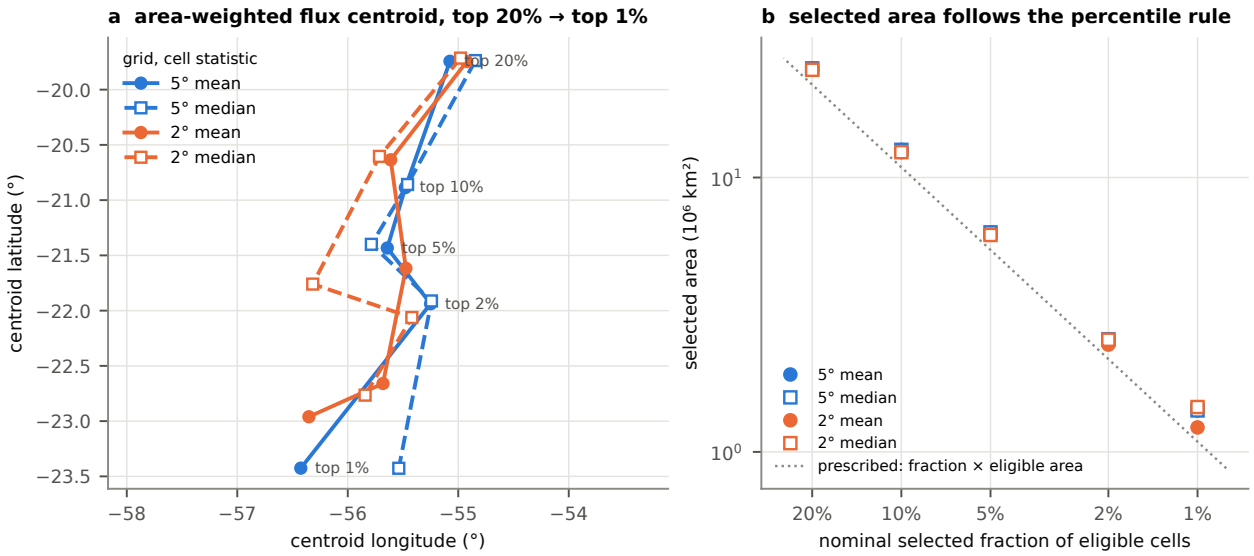


Figure 2: Threshold and estimator sensitivity, NOAA-19 p1, January. (a) Flux × area weighted centroid of the top 20, 10, 5, 2 and 1% footprints for 5° and 2° grids and cell mean (filled) or median (open, dashed). Labels mark the 5° mean path. (b) Selected area against the nominal selected fraction, both on log scales. The dotted line is the area implied by selecting that fraction of the eligible 5° area. Most of the contraction is prescribed by the percentile rule.

3.3 Proton channel

At the reference setting, the p2 and p3 footprint centroids lie south and east of p1 (Fig. 3a,b). The signed differences (second minus first) are −0.891° latitude and +0.673° longitude for p2 − p1 (121.1 km), −1.060° and +1.684° for p3 − p1 (210.4 km), and −0.169° and +1.012° for p3 − p2 (106.1 km). The footprints overlap substantially: the area-weighted Jaccard indices are 0.913 (p1/p2), 0.872 (p1/p3) and 0.955 (p2/p3). For the top-10% footprint over both grids and statistics, the centroid distances between channels range from 83 to 212 km; other thresholds are listed in Supplement S4.

In paired calendar-block resampling, all 1,000 replicates were defined at every block length. The p3 − p1 latitude difference kept its sign throughout the central 95% of replicates: −1.51 to −0.77° for 1-day blocks, −1.46 to −0.87° for 3-day and −1.31 to −0.82° for 7-day. The longitude difference ranged from +0.58 to +2.17°, +0.56 to +2.21° and +0.80 to +2.06°. The p2 − p1 longitude range stays positive but approaches zero (+0.06 to +1.27° for 1-day blocks). The p3 − p2 latitude range crosses zero at every block length (e.g. −0.29 to +0.03° for 7-day blocks). Replicate medians are not identical to the observed differences, and widths change with block length; no block length was selected as preferred. These ranges describe the stability of the p1/p3 ordering under one month of day resampling. They do not remove the shared fitting assumptions linking the channels, the channels' different response functions, or calibration uncertainty.

The continuous normalized fields give a different summary (Fig. 3c). On the common eligible support, which is identical for the three channels (432 cells at 5°, 2,685 at 2°), the p1/p3 mass overlap is 0.933 and pairwise overlaps span 0.919–0.972 over both grids and statistics. The whole-field p1/p3 moment distance is 143.2 km, compared with 210.4 km between footprint centroids. For p2/p3 the whole-field distance is only 42.0 km, and its latitude difference changes sign (+0.120°, versus −0.169° for the footprints). The choice of summary is therefore material: thresholding concentrates weight in the upper tail, and in this case the channels' fields differ most there. The 143.2 km and 210.4 km figures summarize different statistics; their difference is not a causal threshold effect.

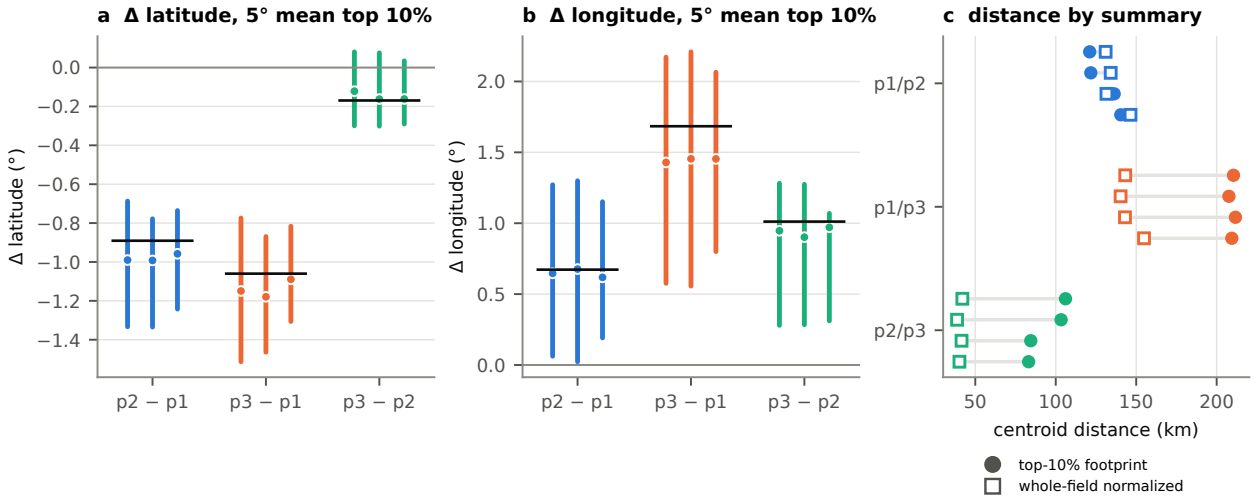


Figure 3: Channel comparison, NOAA-19, January. (a, b) Signed footprint-centroid differences (second channel minus first), 5° mean, top 10%. Black line: observed difference. Coloured bars: 2.5–97.5% range of 1,000 paired calendar-block replicates; within each pair, left to right are 1-, 3- and 7-day blocks, and dots are replicate medians. These are conditional resampling ranges, not confidence intervals for a physical position. (c) Centroid distance between channels for the top-10% footprint (filled) and for the whole-region normalized field of Eq. 4 (open), for four grid/statistic rows per pair (top to bottom: 5° mean, 5° median, 2° mean, 2° median). Absolute fluxes are not compared across channels.

3.4 Accumulation window

One-day maps are sparse: the 1 January NOAA-19 map has 201 eligible 5° cells (44.8% of the regional area), and a median eligible cell was sampled on 1 day, by construction. Eligible area rises to 74.5% for three days and 99.3% for seven (Fig. 4a); at 2° the corresponding fractions are 18.4%, 45.8% and 76.4%.

End to end, the three-day top-10% footprint differs strongly from January: area-weighted Jaccard 0.540, centroid distance 138 km (Fig. 4b,c). Only 61.5% of the January footprint's area is even eligible in the three-day map. Restricted to the 326 cells eligible in both maps, with each map's top-10% recomputed there, the conditional footprints agree at Jaccard 0.829 and 14 km. For one day the end-to-end values are 0.329 and 289 km and the

conditional values 0.823 and 174 km. A substantial conditional difference therefore remains for one day. For seven days, end-to-end agreement is already high (0.887, 39 km); conditioning changes little (0.907, 42 km). Over all 160 non-reference rows (8 windows, counting the week-1 alias, × 20 settings), the shared-support Jaccard index is at least the end-to-end value in 155, but the shared-support centroid distance is **larger** in 38 rows. Conditioning on support does not uniformly improve agreement, and no row was undefined.

Grid resolution matters. At 2° the three-day end-to-end Jaccard index is only 0.336 (conditional 0.837), and the one-day 2° footprint has an end-to-end Jaccard index of 0.130 yet a centroid only 83 km from January's. Centroid distance alone can therefore understate footprint disagreement. The disjoint weeks are mutually consistent at the reference setting: pairwise week-to-week centroid distances are 16–118 km, and the second week is the most distant from January (80 km end to end, 65 km conditional). The 14-day map is nearly identical to January (5 km). This is expected, since it comprises almost half of the month.

These results describe January 2024 sampling by one spacecraft. At the reference setting, most of the three-day end-to-end difference does not persist on shared support, whereas the one-day difference does; they do not identify a universal accumulation time. They also do not separate sampling from genuine day-to-day change: the conditional comparison removes the eligibility difference only, and the month used as reference contains each window.

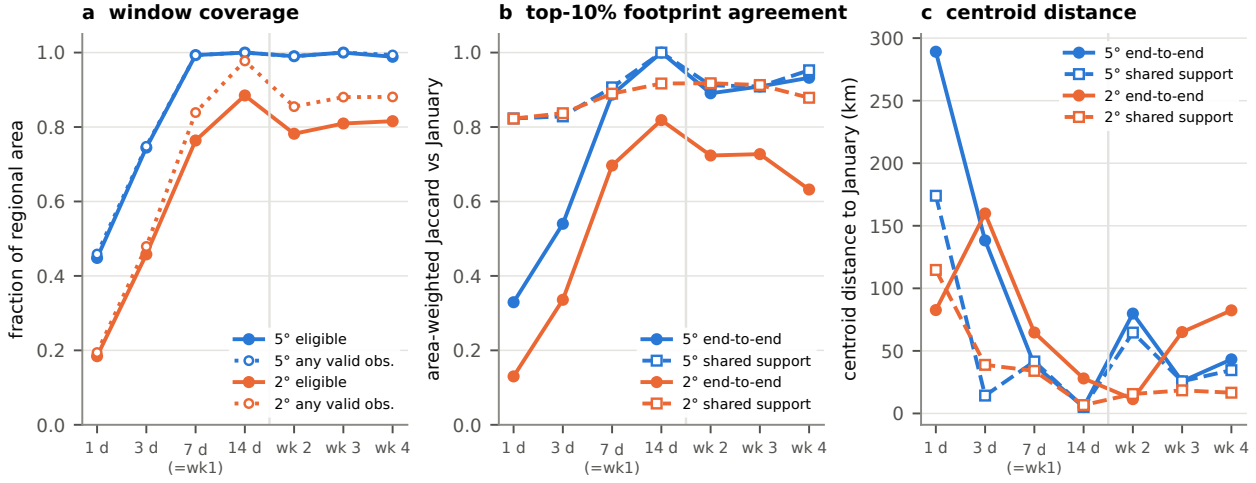


Figure 4: Accumulation-window diagnostics, NOAA-19 p1, cell mean, top 10%, against the January map at the same grid. Cumulative windows (1, 3, 7, 14 days from 1 January) are left of the divider; disjoint weeks are right of it (week 1 equals the 7-day window). (a) Fraction of the nominal regional area that is eligible (solid) or has any valid observation (dotted). (b) Area-weighted Jaccard index of the original footprints (end-to-end, filled) and of footprints recomputed on the cells eligible in both maps (shared support, open). (c) Centroid distances for the same two comparisons. January contains every window, so agreement is descriptive, not independent validation.

3.5 Spacecraft

With QC-1 applied, the five spacecraft give principal centroids within a few degrees of one another (Fig. 5a). The largest separation is 113.4 km (NOAA-15/NOAA-18; 98.3 km at 2°), the NOAA-18 and NOAA-19 centroids are 13.4 km apart, and the largest separation among the four spacecraft other than NOAA-15 is 107.7 km. With cell medians the maximum is 112.5 km. On the 5° mean grid the five-satellite maximum stays between 113 and 157 km over the five thresholds; with medians it reaches 265 km at the top 2% (Fig. 5b), so neither statistic gives the smaller spread at every threshold.

QC-1 excludes one record of the analysed data, the NOAA-15 record of 19 January described in Sect. 2.8, and its exclusion changes NOAA-15 products only. In its 5° cell (centred at −12.5°, −27.5°) the record supplied 88.1% of the summed flux of 526 values. It raised the cell mean from 9.31 to 77.76, and the cell's rank among NOAA-15's eligible cells from 51 to 1, while the cell median (9.34) does not change. In the unfiltered data, that cell carries 8.6% of the NOAA-15 principal centroid weight. The NOAA-15 centroid then lies 225 km further east (−20.96°, −52.89° instead of −21.71°, −54.91°), and the five-satellite maximum is 284.1 km (NOAA-15/MetOp-03; 264.6 km at 2°). The unfiltered mean-based spread also grows with threshold tightness, from 245 to 1460 km, because that cell weighs more as the footprint shrinks. Top-10%/5° footprint membership differs between the two analyses

for 1,015 NOAA-15 regional records. On the 5° grid the median-based maxima are identical with and without QC-1 at every threshold; on the 2° grid they differ by at most 0.001 km.

Neither value is a calibrated inter-satellite difference: the spacecraft differ in local-time sampling and detector history, and a constant intensity factor would not move a percentile footprint or a normalized centroid. Energy-dependent response differences could. Published cross-calibrations for these instruments concern the >16 MeV **integral** channel over earlier decades (Lin et al., 2019). They do not apply to the fitted differential fluxes used here, and we apply none.

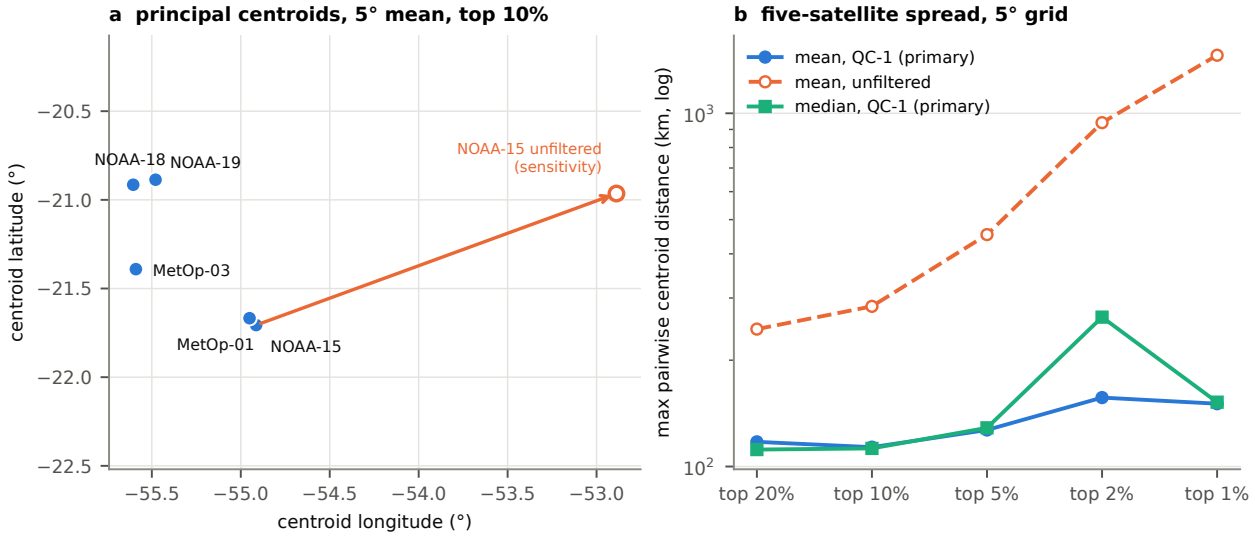


Figure 5: Spacecraft comparison, January 2024, p1, 5° grid. (a) Reference (cell mean, top 10%) centroids of the five spacecraft with QC-1 applied (filled). The arrow points to the NOAA-15 centroid in the unfiltered data, i.e. with the excluded 19 January record included (open; sensitivity; Supplement S7). (b) Maximum pairwise centroid distance among the five spacecraft by threshold (log scale): cell mean with QC-1 (primary), cell mean unfiltered (sensitivity), and cell median with QC-1. The median curve is identical without QC-1 at every threshold (Supplement S7). Intensities are not cross-calibrated; only location is compared.

4 Discussion

4.1 Three different mechanisms

The largest movements reported here arise in different ways, and they should not be ranked against one another. The threshold family moves the centroid by a few hundred kilometres because the percentile **redefines the object**: tighter sets keep fewer, more southerly cells, and their area follows the rule almost exactly. For the top-10% footprint, the channel family moves it by 83–212 km across grids and statistics: the fitted outputs at 25, 50 and 100 MeV place their upper tails in different cells. That ordering held under the day resampling examined here. This study does not separate physical energy structure from the shared fitting model and the channels' response functions. In the unfiltered data, the spacecraft family's largest separation came from **one record** whose flux values do not follow from its own simultaneous >16 MeV count rate. With that record excluded by QC-1, the five spacecraft agree to within 113.4 km at the reference setting, where means and medians give similar values. Short windows add a further effect: they change what can be selected at all.

The ranges of settings are also not commensurate. Five percentiles, three channels, nine windows and five spacecraft were chosen for interpretability, not as samples from comparable distributions of plausible choices. A statement that one factor is "more important" than another would be an artefact of those ranges.

4.2 Implications for particle-based SAA maps

For users of particle-derived SAA locations, four practical points follow. First, report the threshold rule and the population over which it is computed. A percentile over grid cells gives nominal areas by construction, and centroids move non-monotonically as the tail thins. Second, report the eligible support and, when comparing maps, separate end-to-end differences from differences on shared support. Here, most of the three-day end-to-end difference did not persist on shared support, whereas the one-day map still differed there. Reduced support is part of the motivation for multi-spacecraft pooling in short-window studies such as Xu et al. (2025),

although pooling introduces the inter-spacecraft questions discussed above. Third, cell means are exposed to single extreme values. Here one record changed the five-satellite maximum from 113.4 to 284.1 km; a median or a record-level influence check exposes such cases, and a stated rule with the unfiltered result reported alongside keeps any exclusion visible. Fourth, energy channel matters for the thresholded location even where continuous fields overlap substantially.

4.3 Limitations

A footprint centroid is a property of a mapping procedure applied to a sample of fitted fluxes, and it becomes physically interpretable only relative to how that procedure responds to its own choices. The analysis covers one month in one solar and geomagnetic context, so it cannot address seasonal, solar-cycle or secular behaviour of the kind studied in (Fürst et al., 2009; Ginisty et al., 2024; Qin et al., 2014; Schaefer et al., 2016), and the sizes of the sensitivities reported here, including the contrast between one- and three-day windows and the inter-satellite spread, may differ in other months, under other geomagnetic conditions or with other spacecraft. The monthly reference contains every shorter window. Temporal agreement is descriptive, and the paired resampling uses only 31 days, with no established stationarity. Seven-day blocks have 25 possible starts, dates near the month's ends are under-represented, and each join between blocks breaks the dependence between days. The fitted p1–p3 outputs share counts and a spectral model, so channel differences do not come from independent instruments. `mep_IFC_on = −1` is retained without a documented meaning. The processed files carry a provider warning about known contamination, so the data are not certified free of other artefacts. QC-1 was adopted after the influential record had been found, and its parameters were chosen with that knowledge, although the excluded set does not change over wide parameter ranges. It can act only where both neighbourhoods received regular fits, which holds for 32.5% of regional records; elsewhere (at low count rates, near gaps and at file edges) a single-record artefact would not be detected, and artefacts spanning several records are outside its scope. The level-1a evidence shows an internally inconsistent count-rate vector for the excluded record, not why it arose. No cross-calibration is applied: absolute intensities across spacecraft and channels are not compared, and we cannot exclude energy-dependent response differences as a contributor to spacecraft offsets. The IGRF-derived fields in the files are used only descriptively in the supplement. There, footprint samples sit in the lowest modelled field-strength quartile of each spacecraft's region (Supplement S8). This co-location is descriptive: it does not validate a boundary, and a retired legacy classification is not used. Finally, historical bytes of some raw files used in earlier project checkpoints were not recorded. Regional values were reproduced exactly, but those bytes are not reconstructed.

5 Conclusions

For January 2024 NOAA/NCEI SEM-2 omnidirectional proton fluxes, a percentile-defined SAA footprint over South America and the South Atlantic is persistent. Its summary location and extent, however, depend on analysis choices by amounts comparable to the separations one might otherwise interpret. Threshold choice moves the centroid by 351–433 km, with an area change that the rule itself prescribes. For the top-10% footprint, channel choice moves it by up to about 212 km, with the p3-versus-p1 direction persisting under paired day resampling. For three days, most of the end-to-end disagreement with the monthly map does not persist on shared eligible support, whereas a one-day map still differs there. With one NOAA-15 record excluded by a stated post-hoc rule, the five-satellite spread is 113.4 km (112.5 km for cell medians); without the rule it is 284.1 km for cell means. Reporting the selection population, eligible support, estimator and influential observations alongside any particle-derived SAA location would make such maps comparable across studies.

The contribution of this work is that methodology: explicit estimands and denominators, end-to-end and shared-support comparisons, a resampling protocol fixed before production resampling, and record-level influence tracing with a stated quality rule and its unfiltered sensitivity, all regenerated from hashed inputs with values drawn from a claims register. The numbers above are specific to January 2024 and these spacecraft. They establish that such effects can be as large as the separations one might interpret, not how large they are in general; establishing that requires applying the same procedure to other months and conditions.

6 Data and code availability

Input data are the public NOAA/NCEI processed SEM-2 daily files (NOAA National Centers for Environmental Information, 2024) for 1–31 January 2024 (five spacecraft); file names and SHA-256 hashes are listed in the supplement's input manifest. One level-1a companion file from the same archive (`poes_n15_20240119_raw.nc`) was read only as evidence for the excluded record; its hash is listed there too. Analysis code, the frozen resampling protocol, derived tables, the claims register, figure-generation code and these document sources are in the project repository `fellipegoncalvesleite/saa-poes-mapping`. A single command (`python scripts/reproduce_paper.py`) regenerates every derived table, the resampling, the diagnostics, the figures, the claims register, this document and its supplement from the hashed raw files. The version described here is archived on Zenodo as release v1.0.0, doi:10.5281/zenodo.23324359 (Gonçalves Leite, 2026).

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8 Conflict of Interest Disclosure

The author declares there are no conflicts of interest for this manuscript. This research received no specific funding.

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